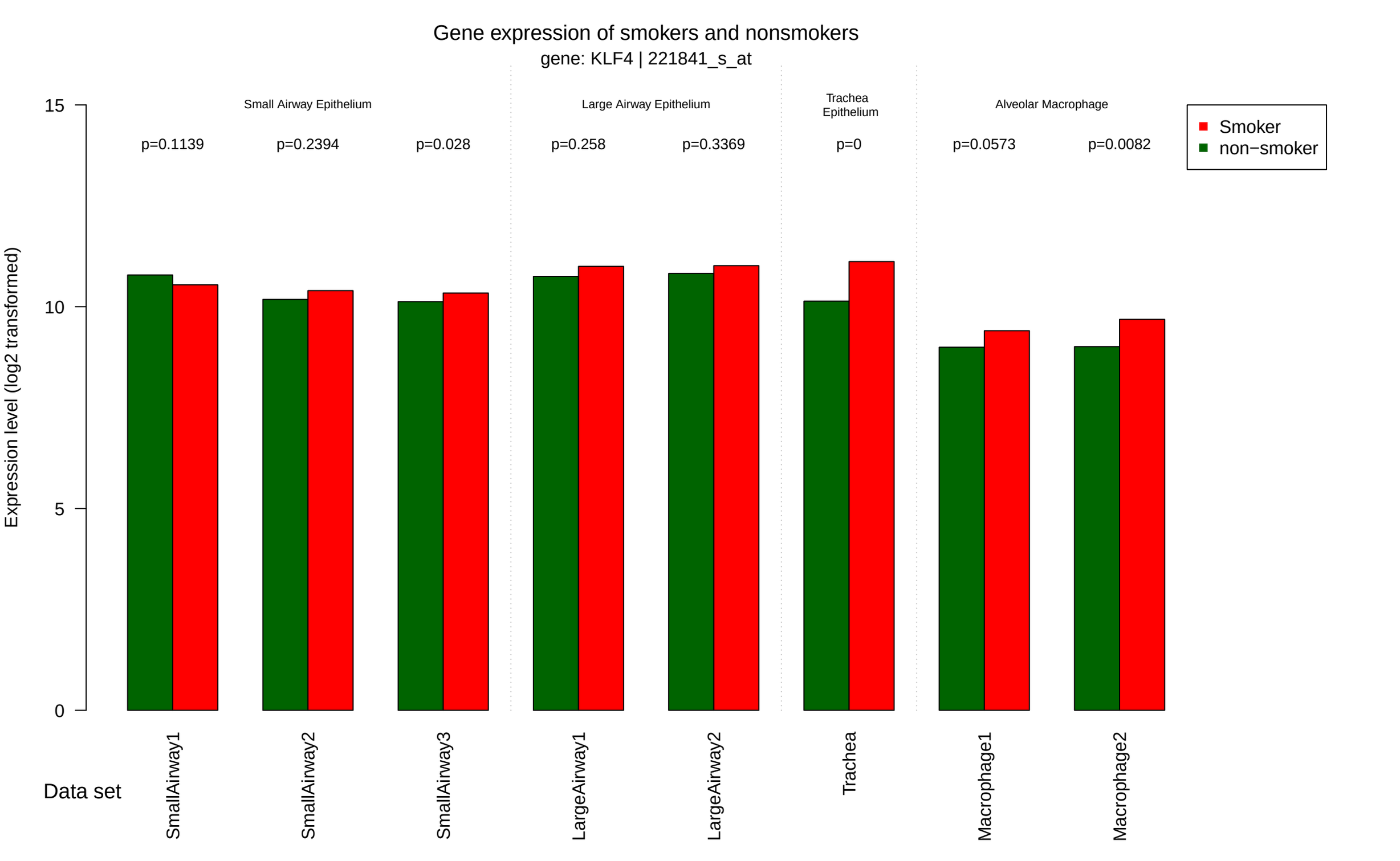
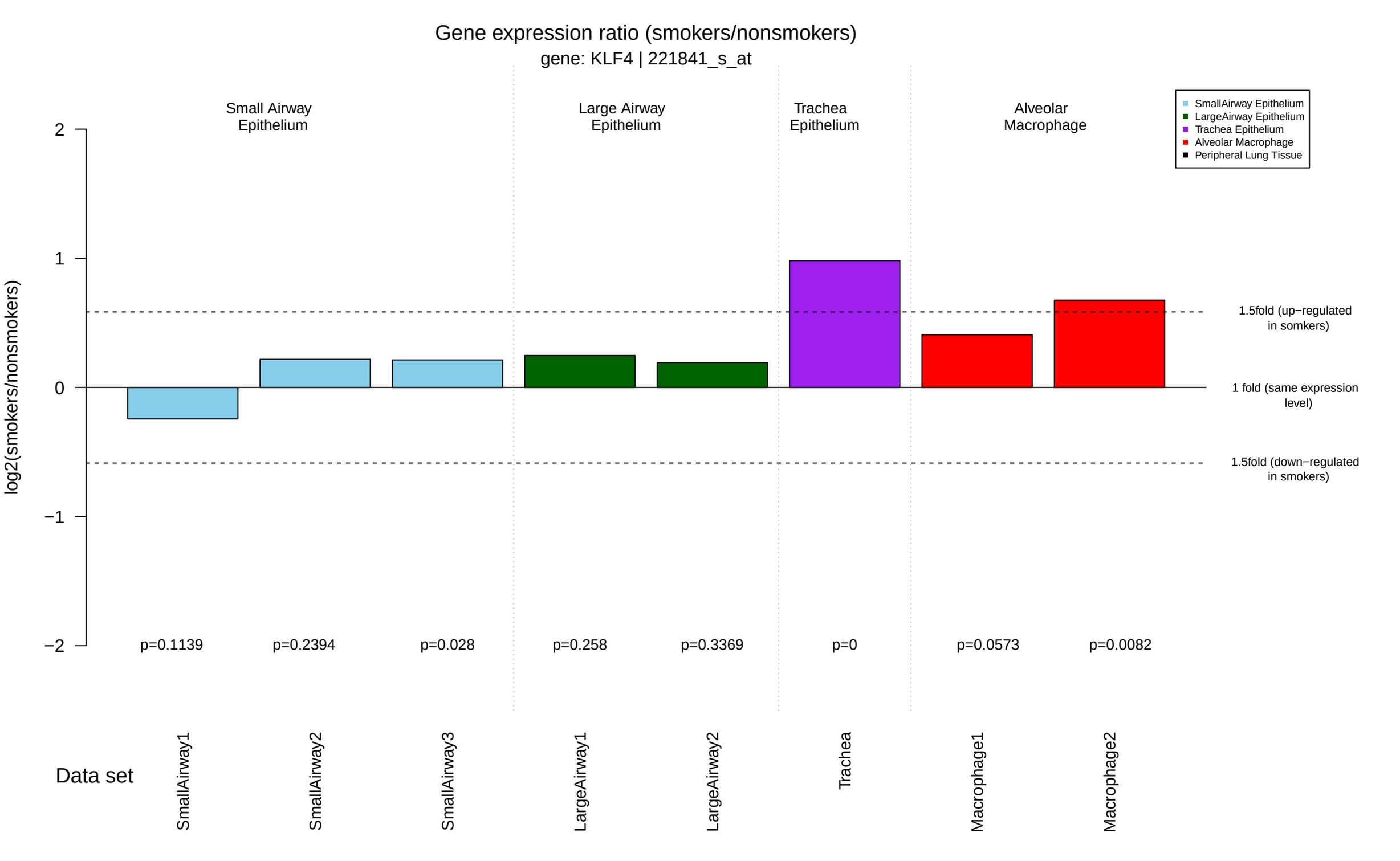
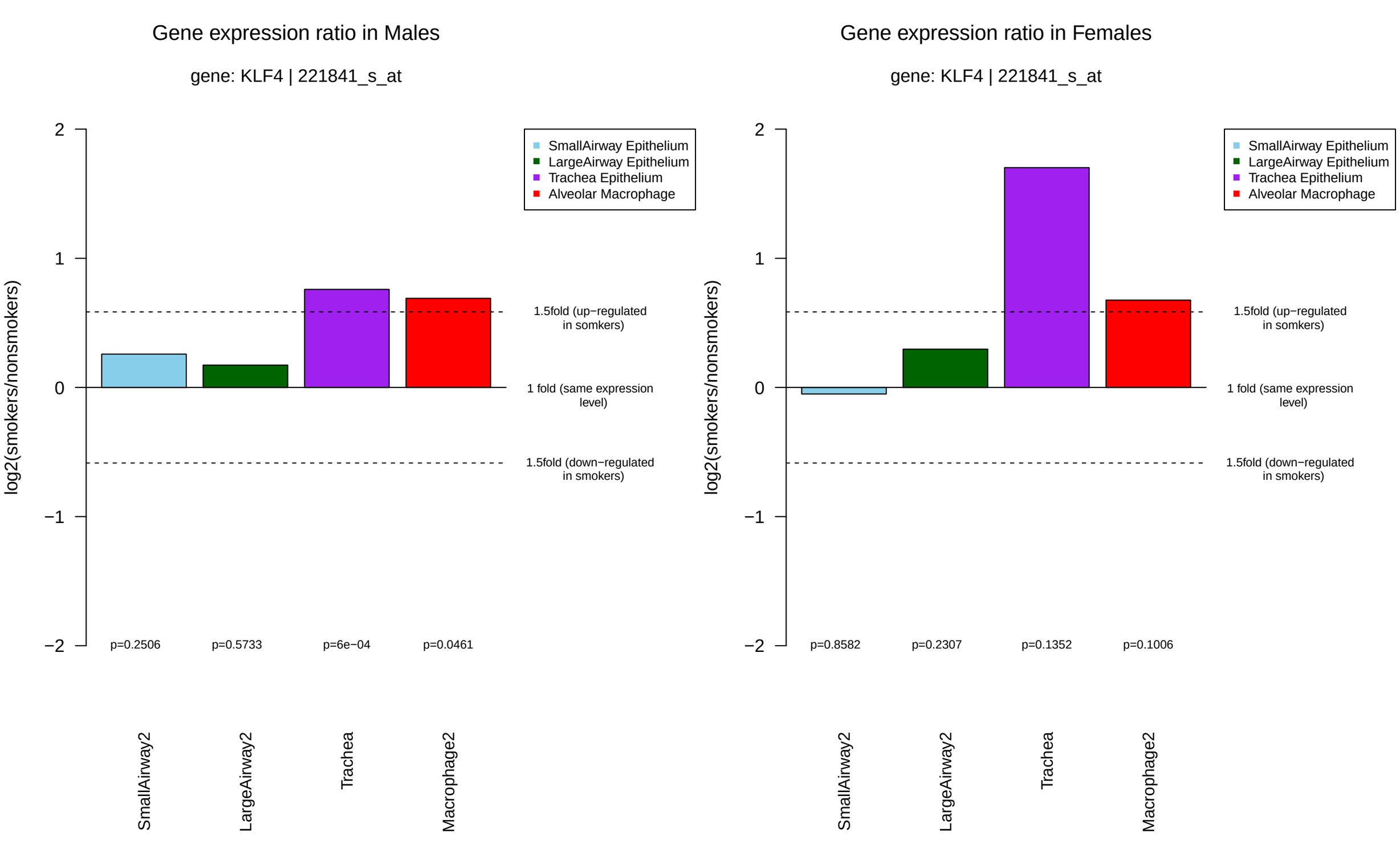
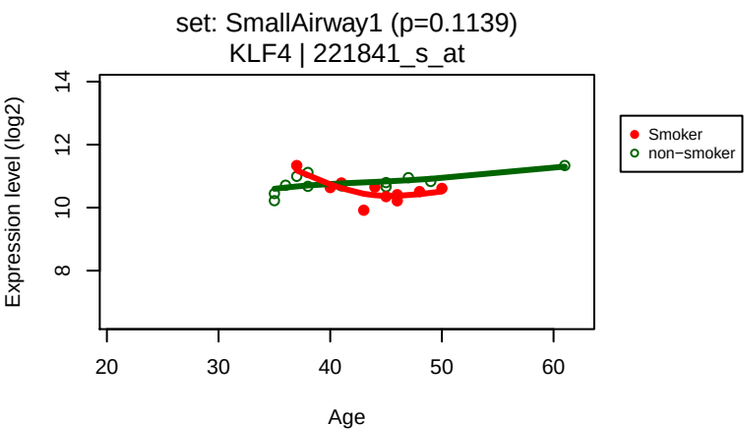
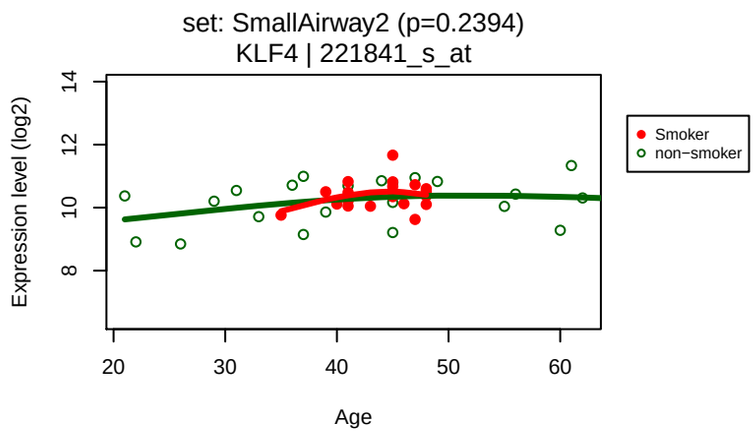
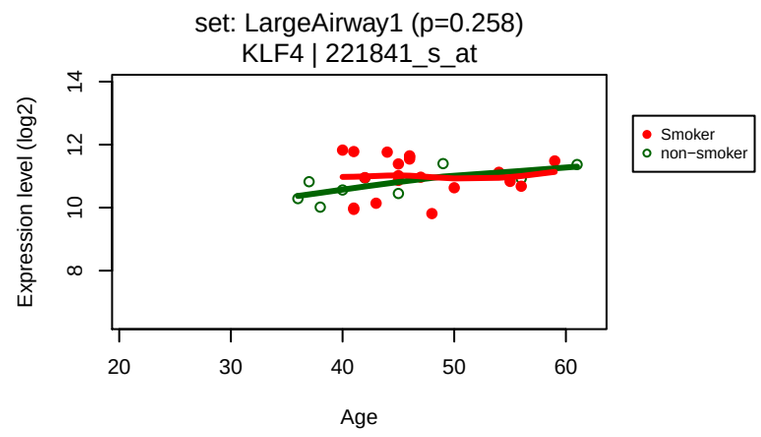

Gene expression of smokers and nonsmokers
gene: KLF4 | 221841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1139
p=0.2394
p=0.028
p=0.258
p=0.3369
p=0
p=0.0573
p=0.0082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

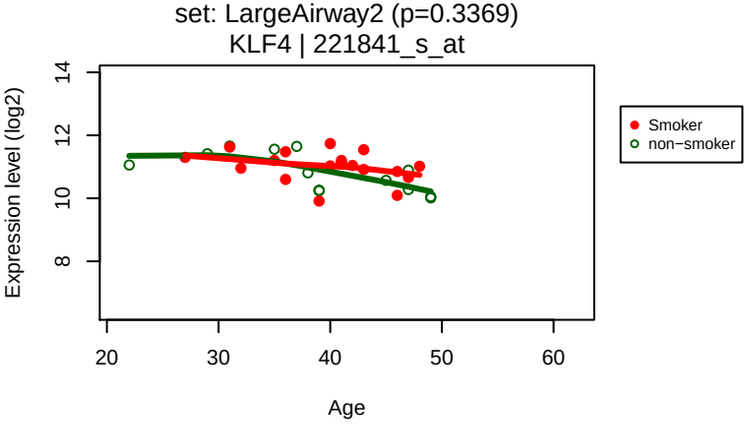

Gene expression ratio (smokers/nonsmokers)
gene: KLF4 | 221841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1139
p=0.2394
p=0.028
p=0.258
p=0.3369
p=0
p=0.0573
p=0.0082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

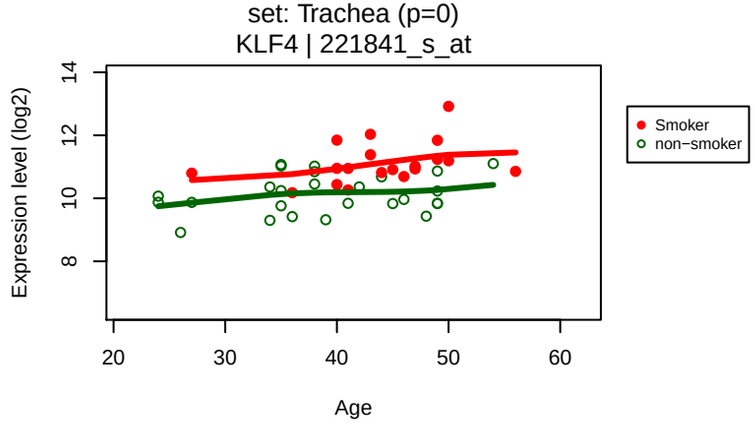

Gene expression ratio in Males
gene: KLF4 | 221841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2506
p=0.5733
p=6e−04
p=0.0461
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLF4 | 221841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8582
p=0.2307
p=0.1352
p=0.1006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

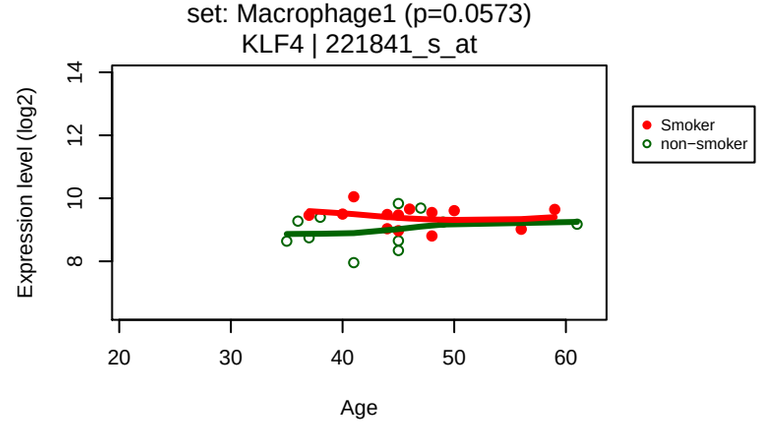

set: SmallAirway1 (p=0.1139)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker

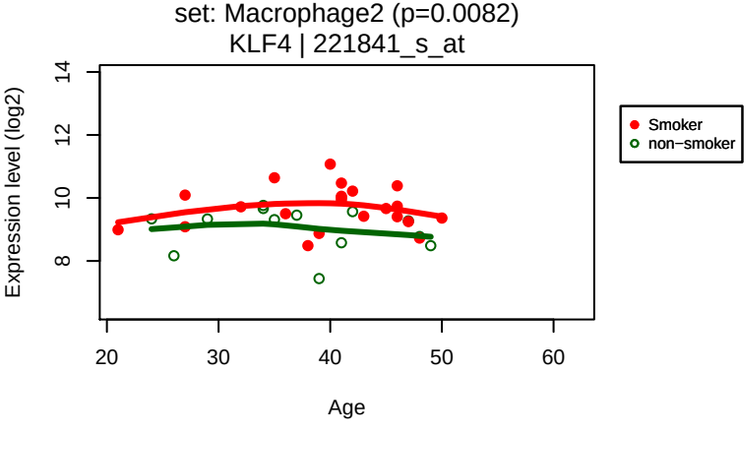

set: SmallAirway2 (p=0.2394)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker

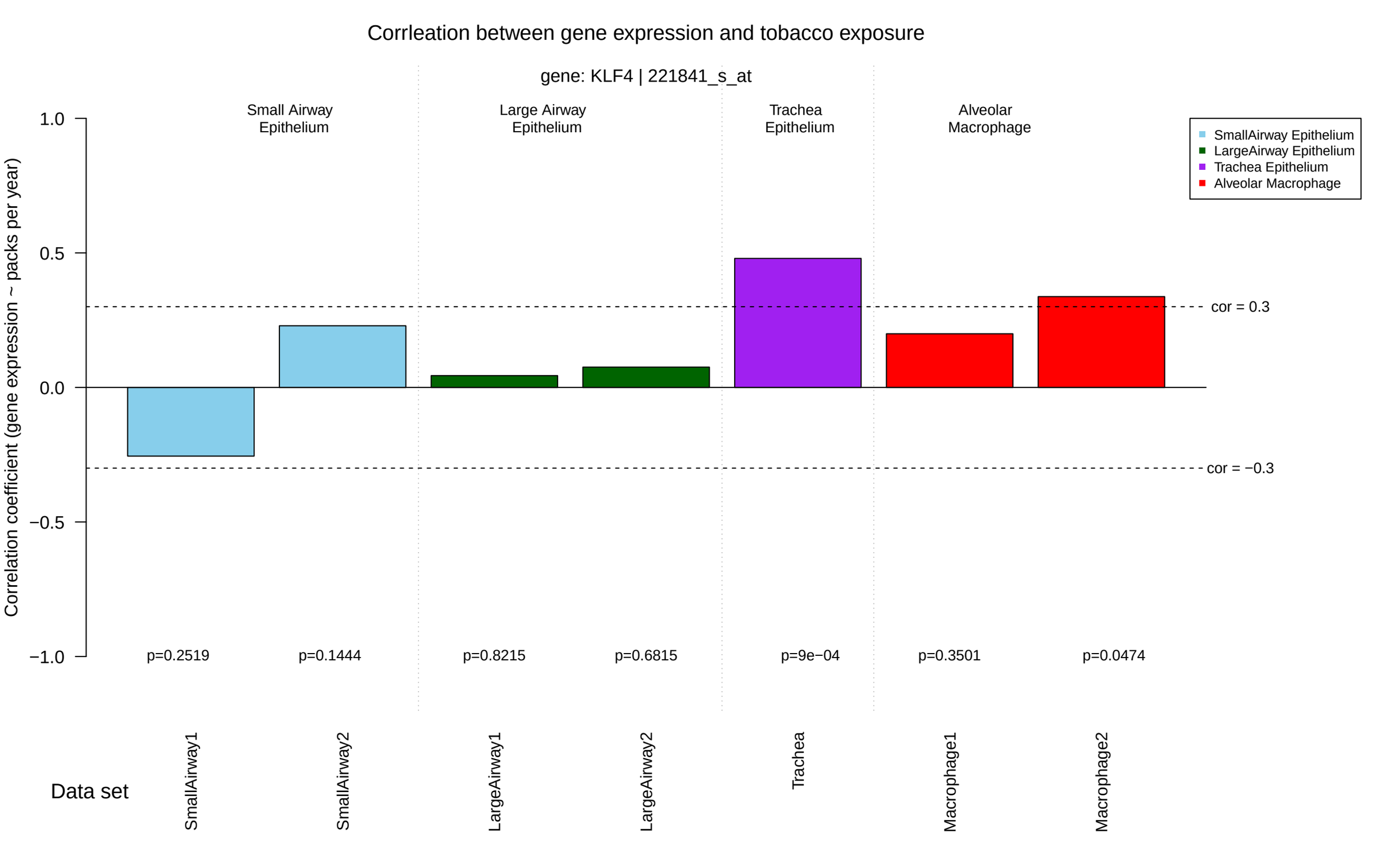

set: LargeAirway1 (p=0.258)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3369)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0573)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0082)
KLF4 | 221841_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLF4 | 221841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2519
p=0.1444
p=0.8215
p=0.6815
p=9e−04
p=0.3501
p=0.0474
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2